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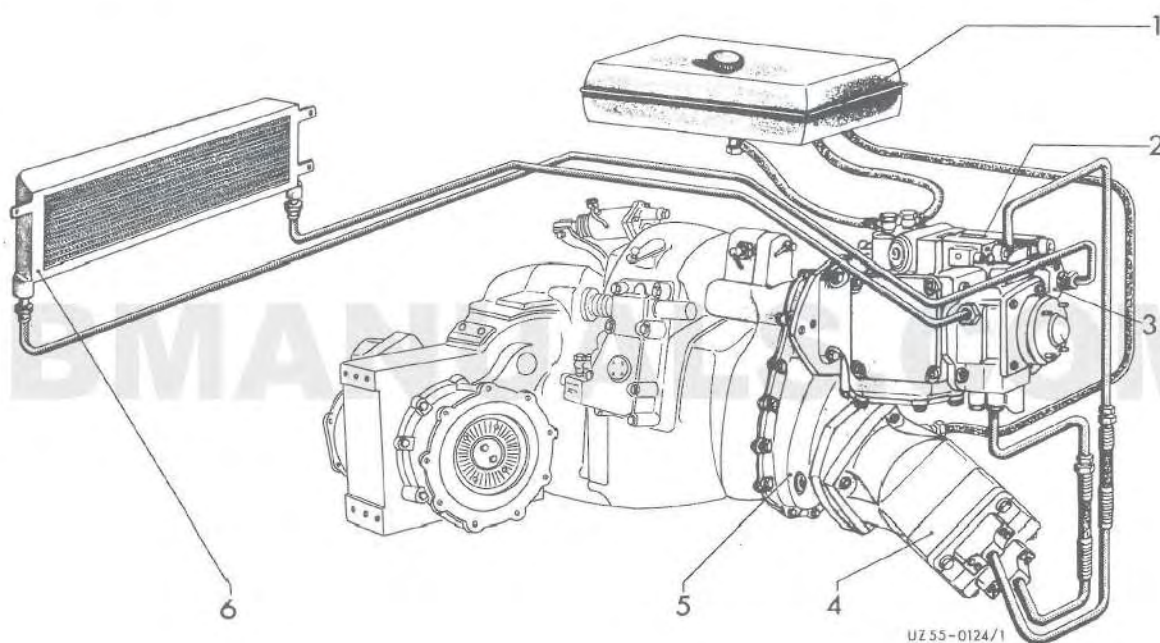
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Installation survey

Chassis Model	Sales designation	Hydrostat manufacturer	Version	Installation
435,110	U 1700	Linde Ltd.	Variable hydropump and constant hydromotor	Special 36 036
.111	U 1700 L ¹⁾			

1) L = Lorry

General view



Hydrostat

- | | |
|------------------|-----------------------------|
| 1 Expansion tank | 4 Hydromotor |
| 2 Hydropump | 5 Intermediate transmission |
| 3 Oil filter | 6 Oil cooler |

Technical details**Hydropump**

Version		Variable hydropump with oil filter
Manufacturer		Linde
Designation		BPV 70S
Design		Swash plate, closed circuit
Max. permissible speed	rpm	3000
Max. delivery	per rev.	70.9 cm ³
Max. pressure	bar	500
Max. operating pressure	bar	420
Sustained pressure	bar	250

Hydromotor

Version		Constant hydromotor
Manufacturer		Linde
Designation		BMF 75 TF
Design		Eccentric shaft
Max. permissible speed	rpm	3700
Consumption rate	per rev.	75 cm ³
Max. pressure	bar	500
Operating pressure	bar	420
Sustained pressure	bar	250

Adjustments**Intermediate transmission**

Permissible end float in the input shaft	mm	0.03 ± 0.02
Permissible end float in the layshaft	mm	0.06 ± 0.02

Capacities

Component	Service product (sheet No. of MB service product specifications)	SAE class	Capacity (L)
Hydrostat system (complete capacity)	Hydraulic oil (341)	—	17.0
	Engine oil (236.2)	10 W ¹⁾	
Main transmission with intermediate transmission	Transmission oil (226)	80 ²⁾	11.5

1) In cold regions SAE 5 W–20/30.

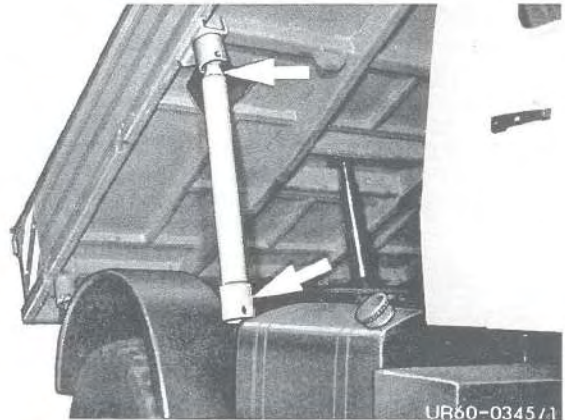
2) Or multigrade oil SAE 80 W 85 W, SAE 85 W 90, SAE 90.

Consumibles

Cons. No.	Description	Part No.
1	Loctite No. 573	001 989 45 20

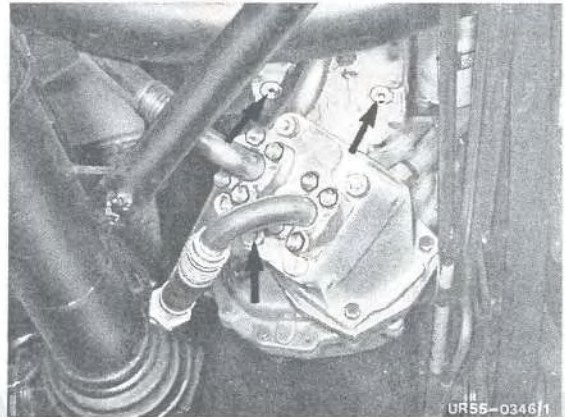
Removal

- 1 Raise platform body and fit safety stay.
- 2 Remove spare wheel.

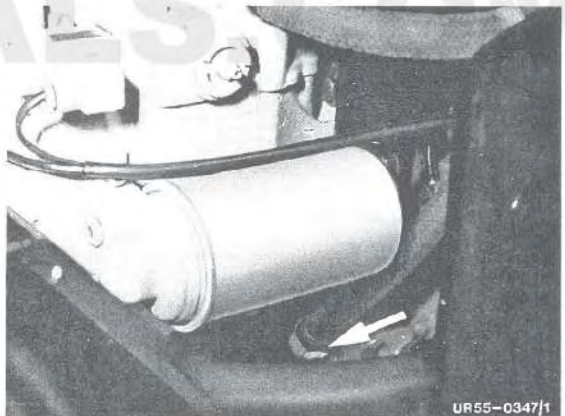


- 3 Drain and retain oil in hydromotor and pump.

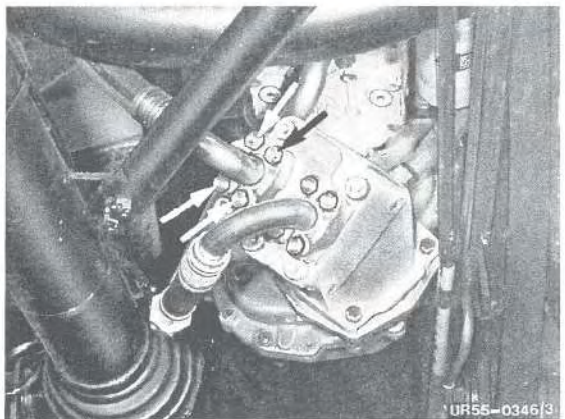
Note: Open cap on expansion tank.

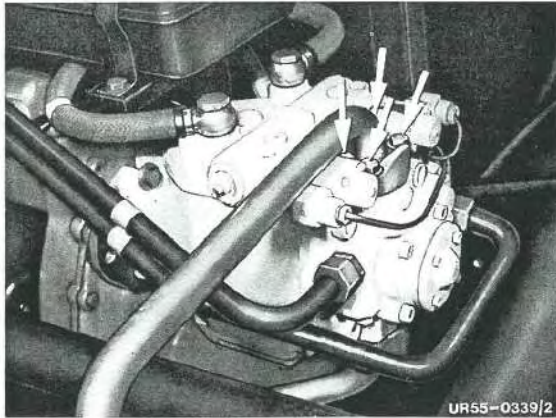


- 4 Remove oil leak-off pipe from hydromotor.

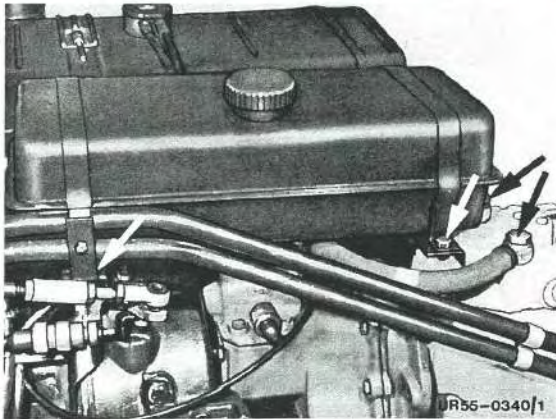


- 5 Remove pressure line to hydromotor.
- 6 Fit new washers to drain plugs on hydromotor and pump.

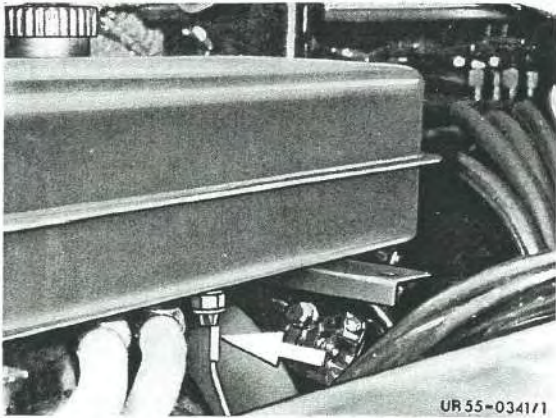




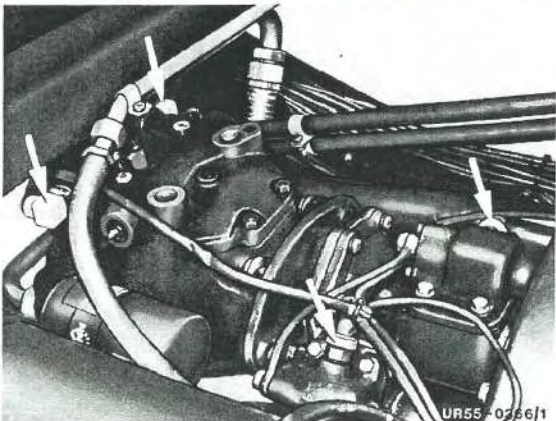
- 7 Remove pressure line from hydropump.



- 8 Remove expansion tank from bracket. Remove supply and return lines to hydropump.



- 9 Disconnect oil temperature sender unit from expansion tank and remove expansion tank.



- 10 Mark cable connectors to the selector cylinders and the hydropump and remove.
- 11 Release loom holder and move loom aside.

12 Remove supply and return lines to the oil cooler from the hydropump.

13 Remove main transmission.
See 26.13-2.1/1 or 2.2/1.

14 Clean all parts, check and change if necessary.

Note: The hydropump and hydromotor are repaired by Linde Ltd.

Installation

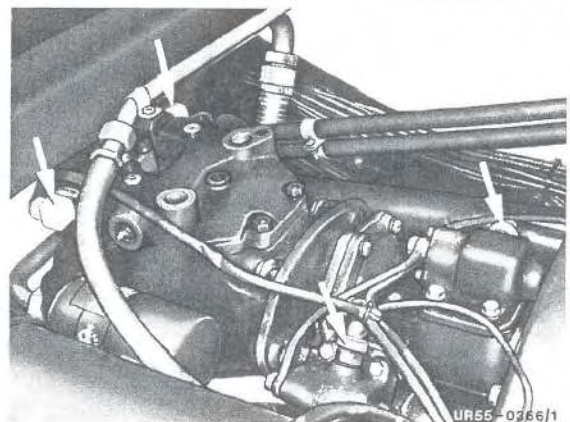
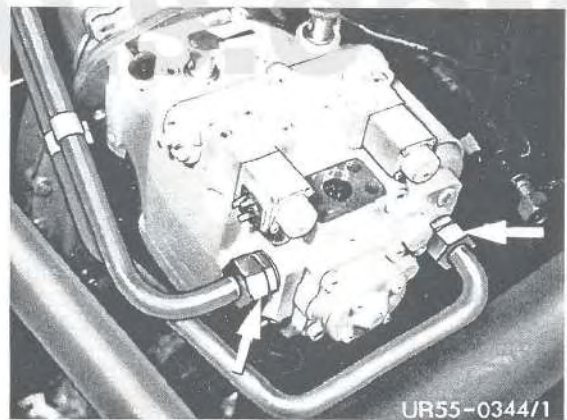
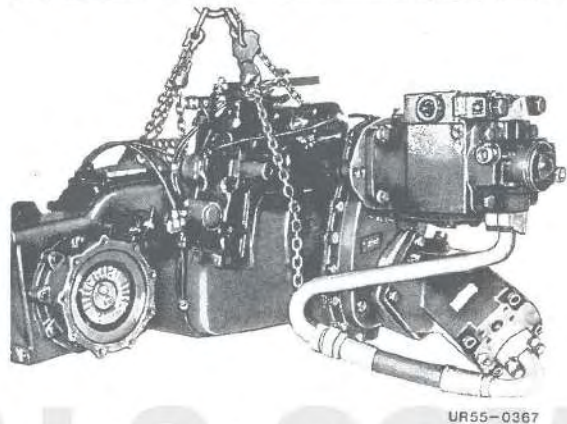
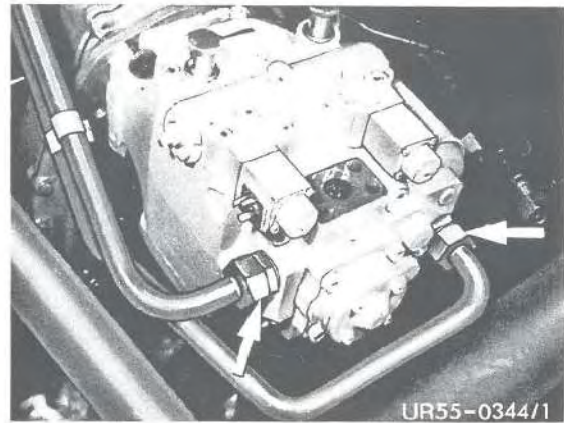
1 Bring main transmission and hydrostat into position with lifting tackle.

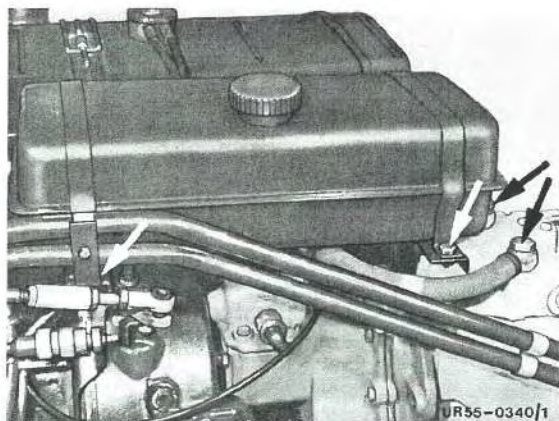
2 Install main transmission,
see 26.13-2.1/1.

3 Fit supply and return lines to the oil cooler with new seals on hydropump.

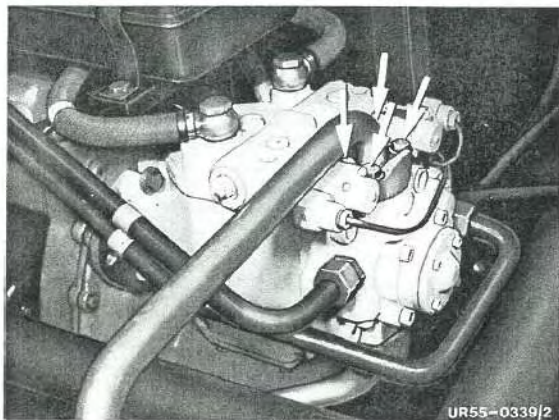
4 Fit cable connectors following markings to the selector cylinders and mount on hydropump.

5 Fasten loom to holder.



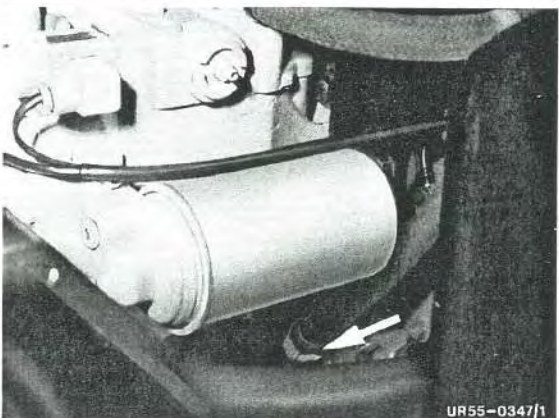


- 6 Fit expansion tank with bracket and attach supply and return lines to hydropump, connect oil temperature sender unit to expansion tank.



- 7 Fit pressure lines with new seals and bolts to hydromotor and pump.

Note: Only use self-locking bolts **once**.

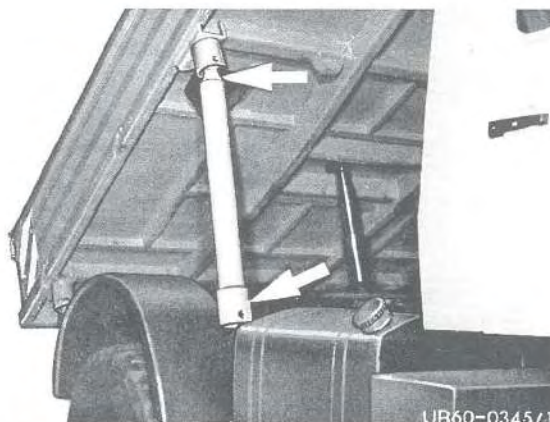


- 8 Fit oil leak-off pipe to hydromotor.

- 9 Fill expansion tank with oil.
10 Bleed hydrostat oil circuit.
11 Fit spare wheel.
12 Lower platform body.

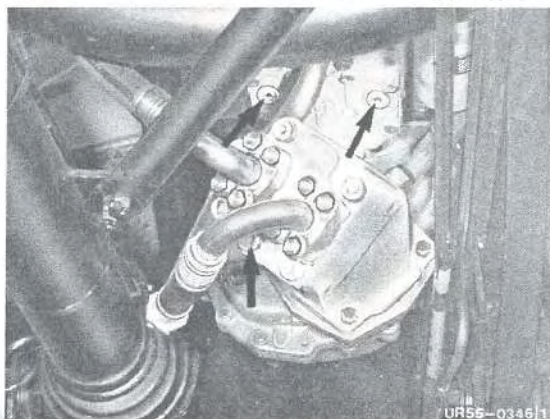
Removal

- 1 Raise platform body and fit safety stay.

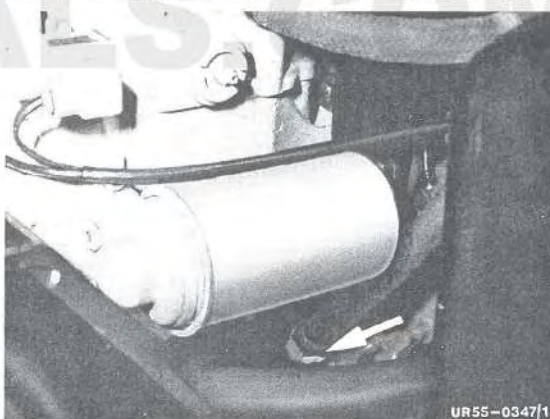


- 2 Drain and retain oil from hydromotor and pump.

Note: Open cap on expansion tank.

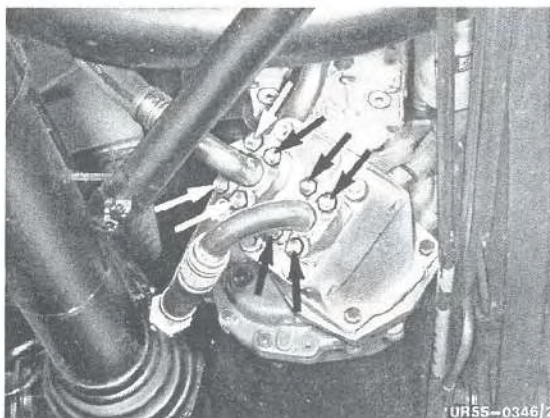


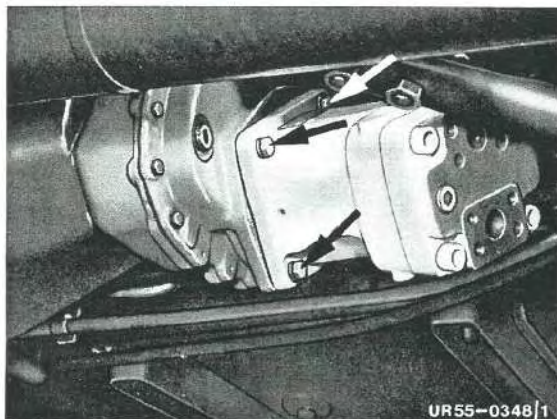
- 3 Remove oil leak-off pipe from hydromotor.



- 4 Remove pressure line to hydromotor.

- 5 Fit new washers to drain plugs on hydromotor and pump.

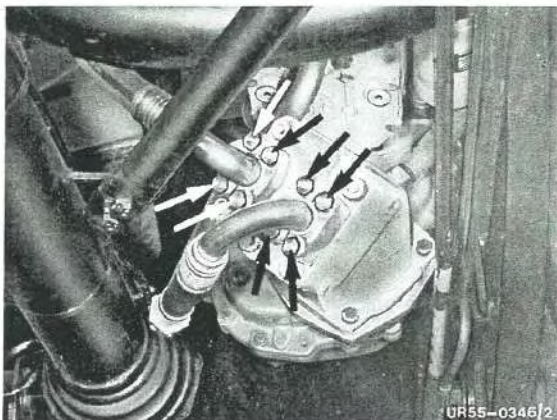




6 Remove hydromotor from intermediate transmission.

7 Clean all parts, check and change if necessary.

Note: The hydromotor is repaired by Linde Ltd.

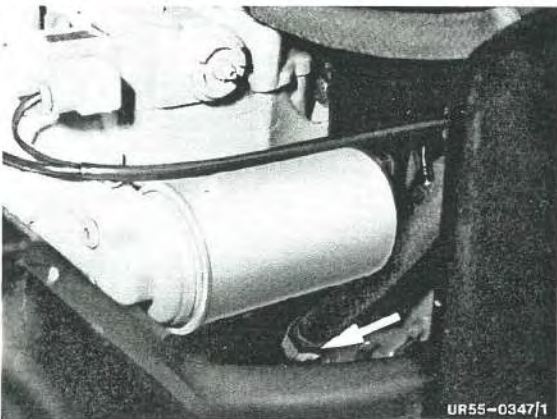


Installation

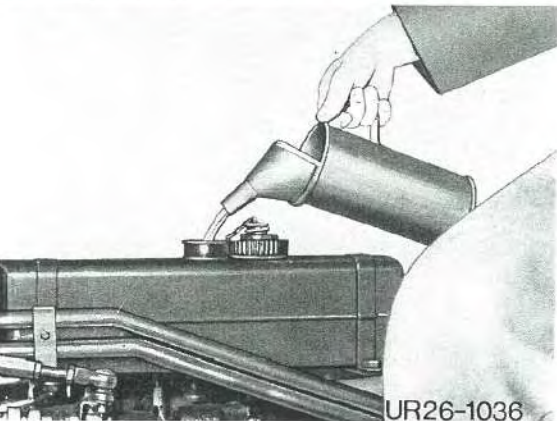
1 Install hydromotor to intermediate transmission.

2 Fit pressure lines with new seals and bolts to hydromotor.

Note: Only use self-locking bolts **once**.



3 Fit oil leak-off pipe to hydromotor.

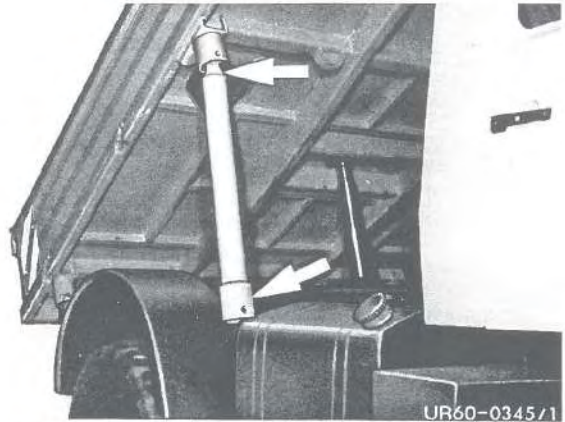


4 Fill expansion tank with oil.

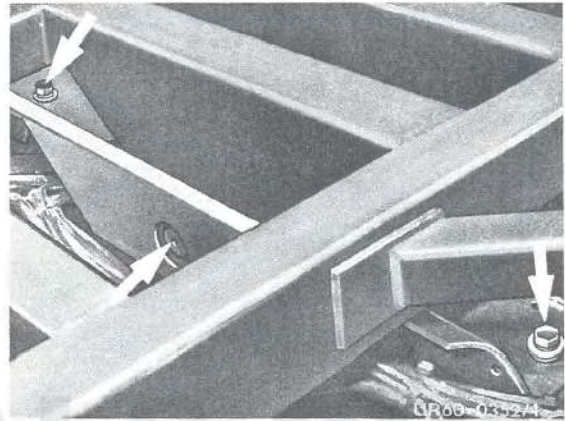
5 Bleed hydrostat oil circuit, see 7.1/1.

Removal

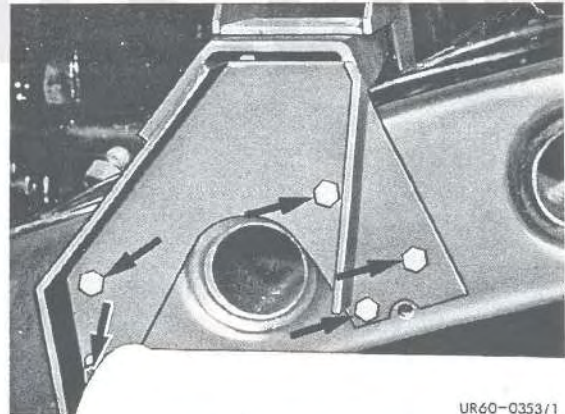
- 1 Raise platform body and fit safety stay.



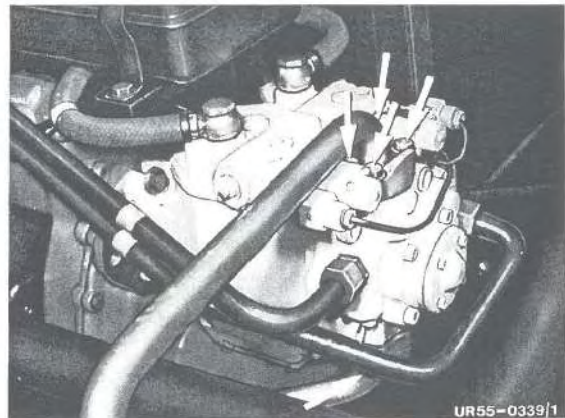
- 2 Loosen cross-member on the frame and sub-frame.

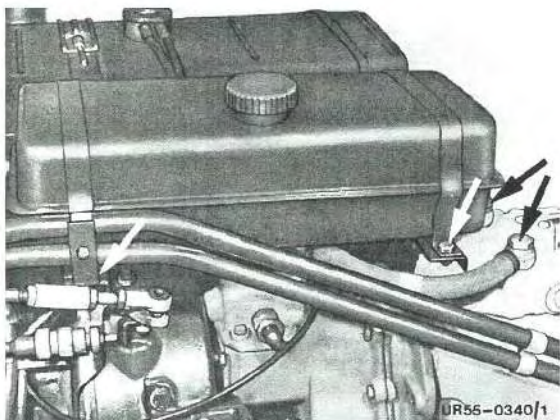


- 3 Remove bracket on the frame.
- 4 Support sub-frame and remove cross-member.

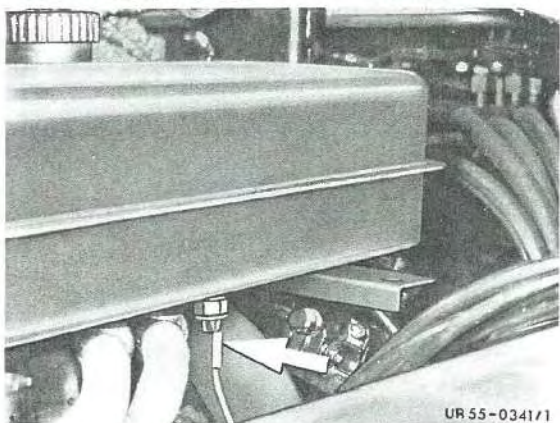


- 5 Remove hydromotor, see 3.1/1.
- 6 Remove pressure lines from hydropump.

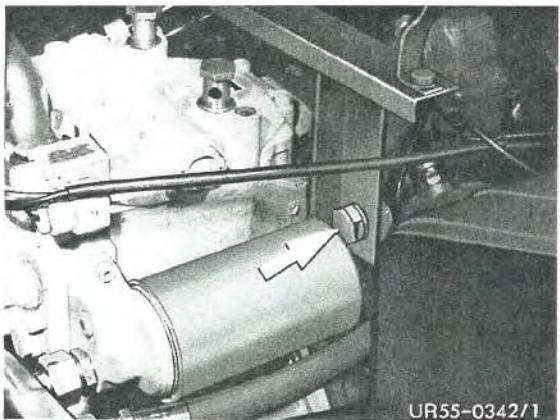




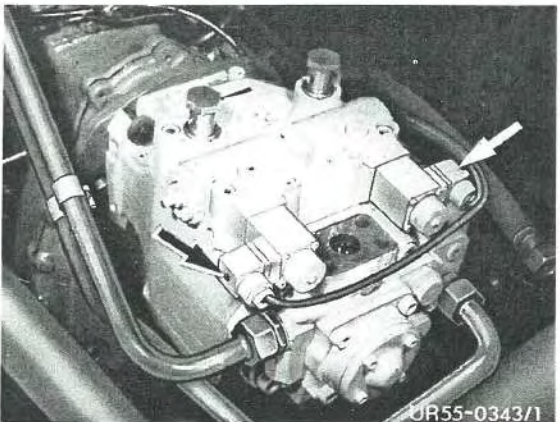
- 7 Remove expansion tank from bracket.
- 8 Remove supply and return line to hydropump.



- 9 Disconnect oil temperature switch on expansion tank and take off expansion tank.



- 10 Remove bracket on intermediate transmission.



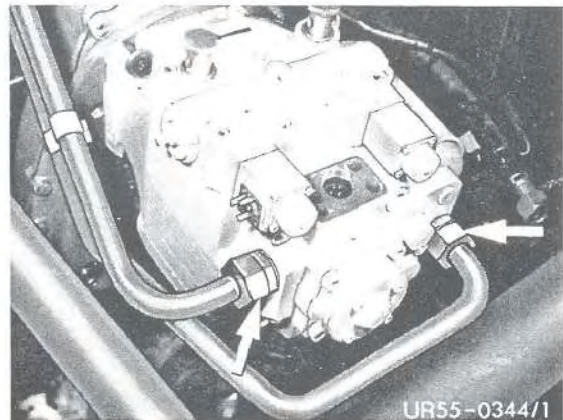
- 11 Mark cable connectors on left and right of hydropump and remove.

12 Remove supply and return line to the oil cooler from hydropump.

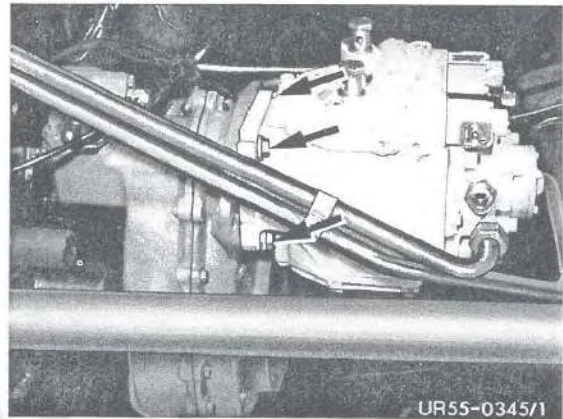
13 Remove hydropump from the intermediate transmission.

14 Clean all parts, check and change if necessary.

Note: The hydropump is repaired by Linde Ltd.



UR55-0344/1

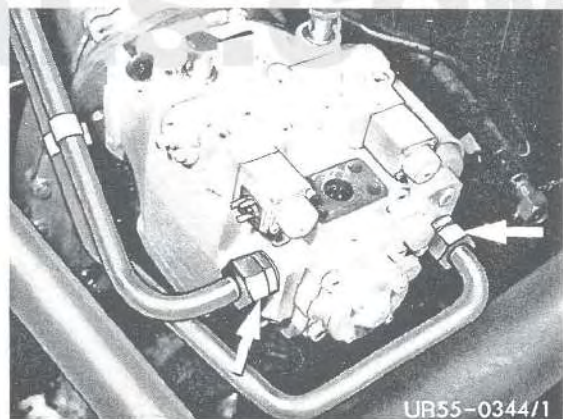


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Installation

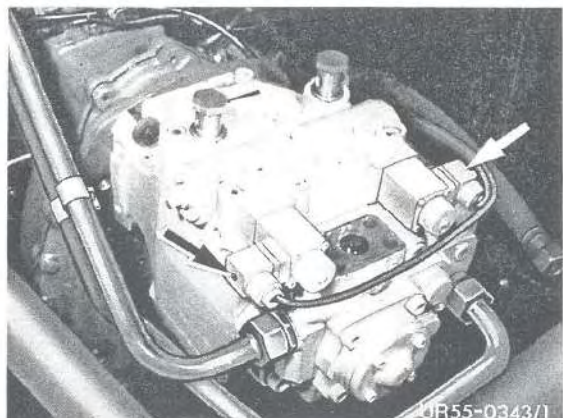
1 Fit hydropump to the intermediate transmission.

2 Fit supply and return line to oil cooler with new seals at hydropump.

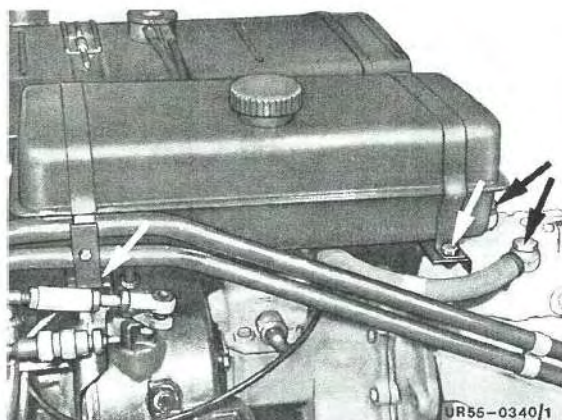


UR55-0344/1

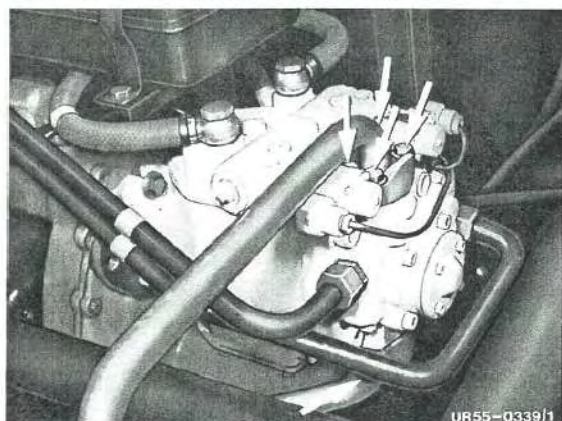
3 Fit cable connectors according to markings on left and right of hydropump.



UR55-0343/1

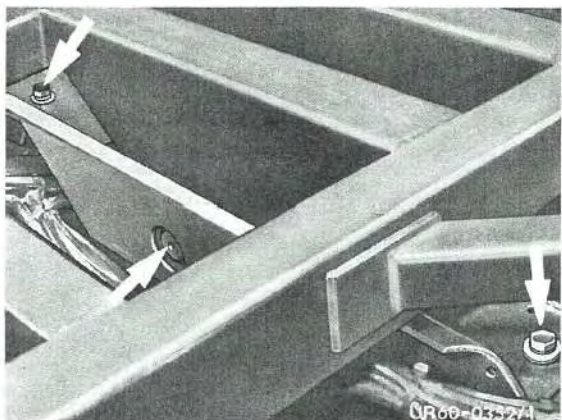


- 4 Fit expansion tank with bracket and supply and return line to hydropump, connect oil temperature sender unit to the expansion tank.

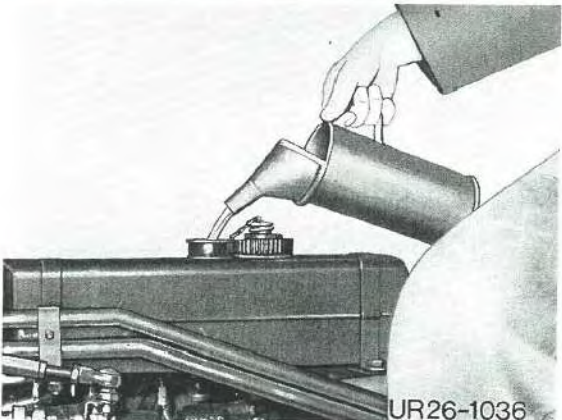


- 5 Install hydromotor, see 3.1/1.
- 6 Fit pressure lines with new seals and bolts to hydromotor and pump.

Note: Only use self-locking bolts **once**.



- 7 Insert cross-member and fit bracket to the frame.
- 8 Fasten cross-member to frame and sub-frame.



- 9 Fill expansion tank with oil.
- 10 Bleed hydrostat oil circuit, see 7.1/1.
- 11 Lower platform body.

Removal

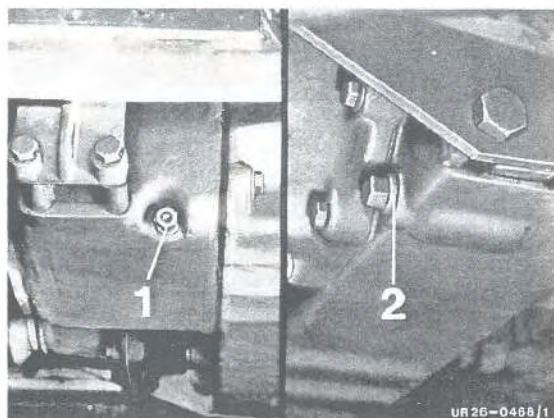
- 1 Remove hydropump, see 3.1/1.
- 2 Drain oil from planetary gearing (1) and transfer box (2).

Note: The main transmission oil circuit is connected to the intermediate transmission.

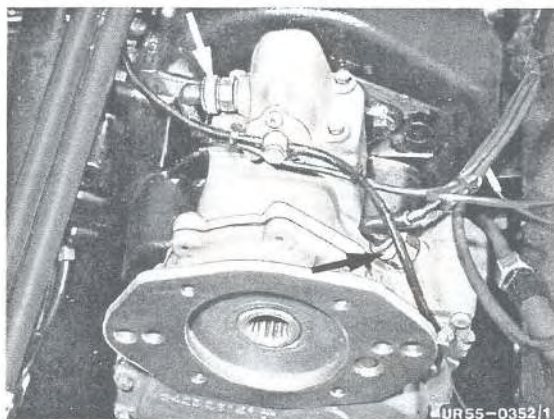
- 3 Remove cable connector from the selector cylinders.

- 4 Remove selector cylinders.

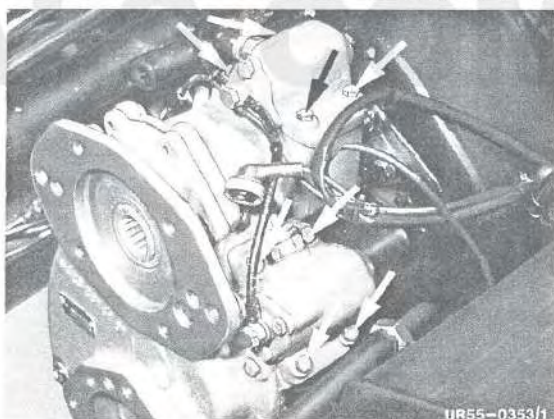
- 5 Remove air lines to the selector cylinders and take off selector cylinders.



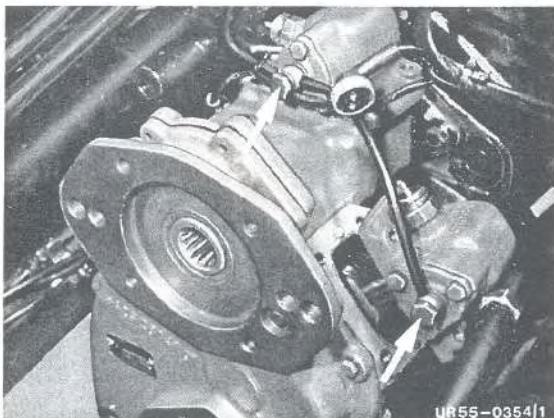
UR 26-0488/1



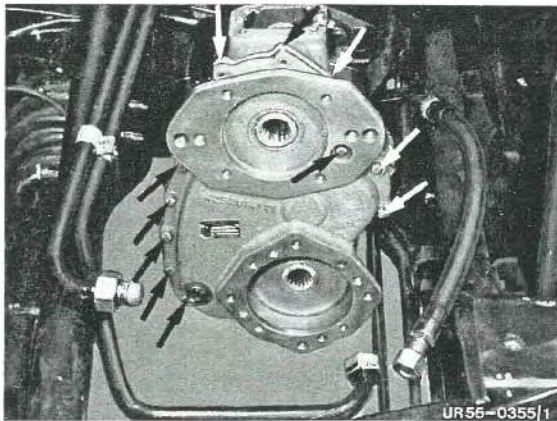
UR55-0352/1



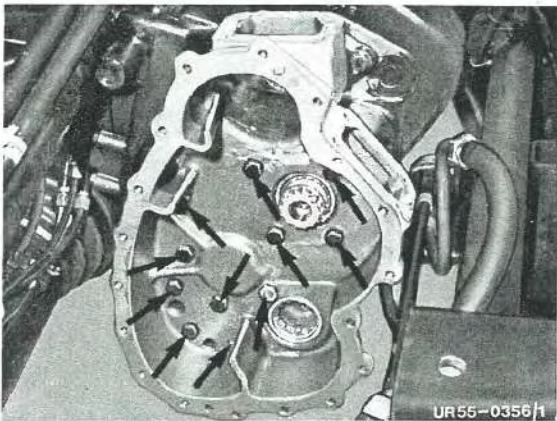
UR55-0353/1



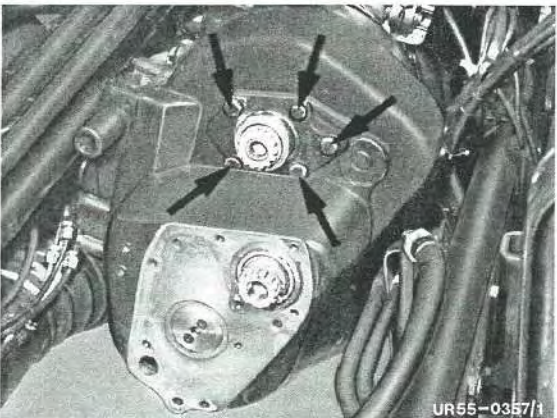
UR55-0354/1



6 Remove intermediate transmission cover with gear train.



7 Remove intermediate transmission casing.



8 Remove circlip from input shaft and take off bearing cap.

9 Clean all parts, check and replace if necessary.

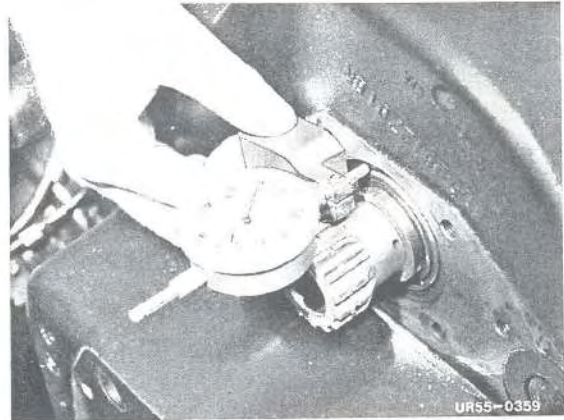
Installation

1 Measure new bearing cap and with it measure stand-proud "A".

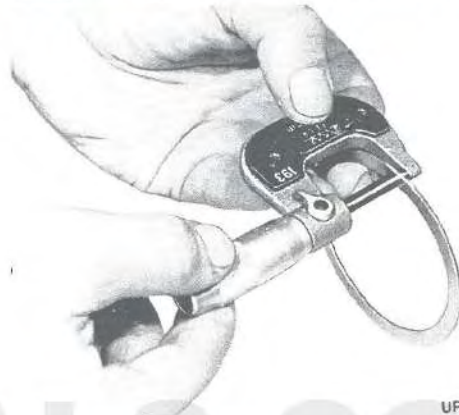


2 Measure stand-down "B" on the input shaft bearing.

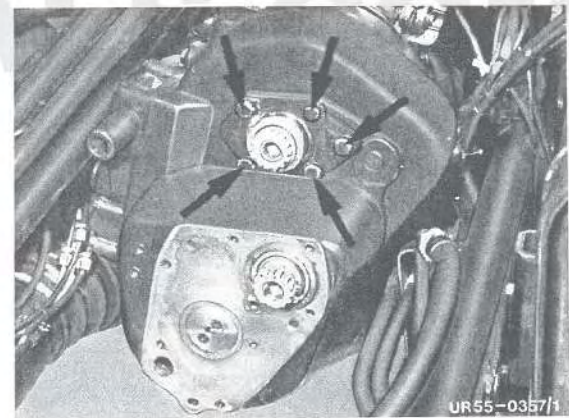
3 Determine the value "C" of the shim.
 $C = (B + \text{gasket thickness}) - (A + \text{bearing play})$.
 Bearing play, see 1.2/1.



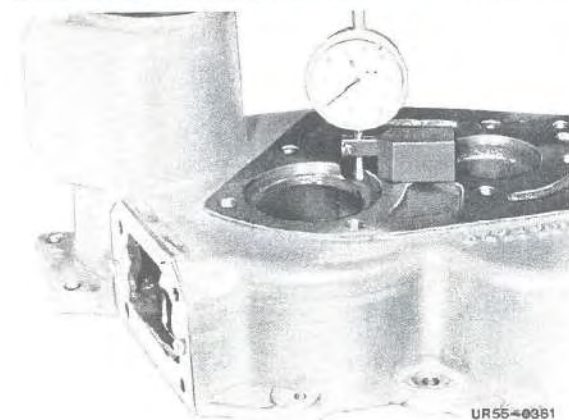
4 Choose shim corresponding to value "C" and fit.

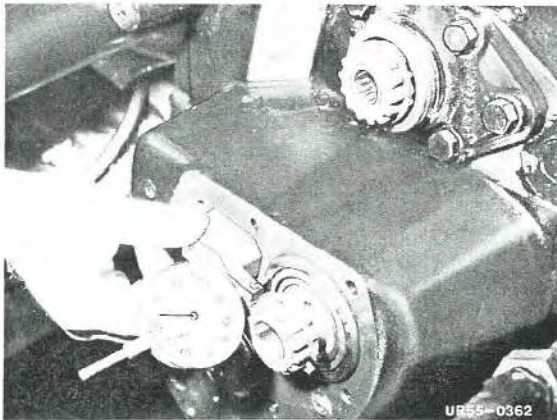


5 Fit bearing cap with new gasket. Screw in bolts with sealant No. 1.



7 Measure new intermediate transmission casing and with it measure the bearing stand-down "A" in the casing.



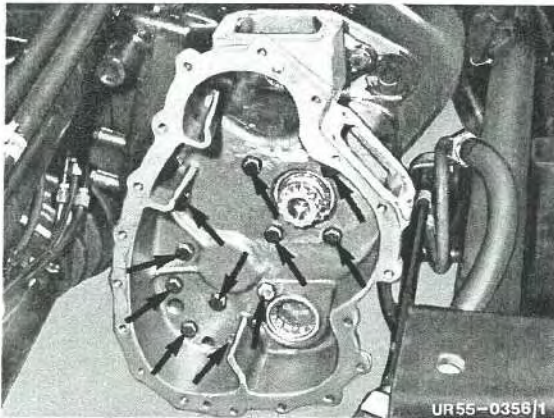


8 Measure bearing stand-proud "B" in the main gearbox.

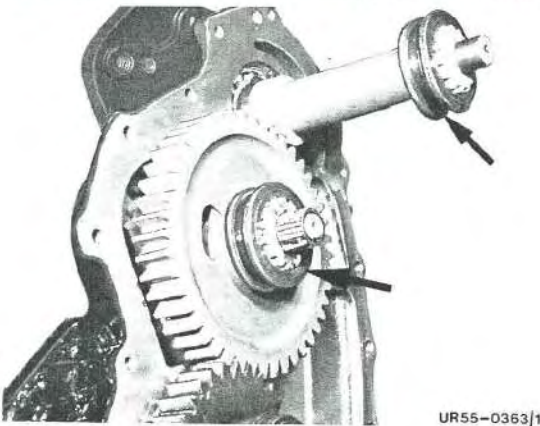
9 Determine the value "C" of the shim.
 $C = (B + \text{gasket thickness}) - (A + \text{bearing play})$.
 Bearing play, see 1.2/1.



10 Choose shim corresponding to value "C" and fit.



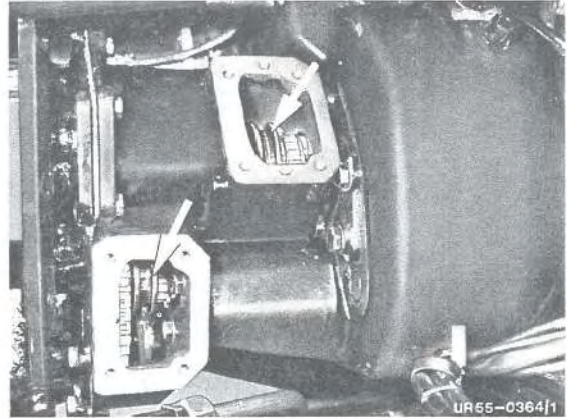
11 Install intermediate casing with new gasket.



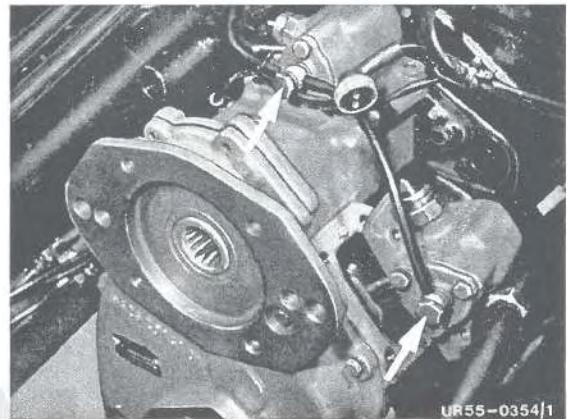
12 Fit sliding sleeves for hydropump input shaft and main transmission primary shaft.

- 13 Install intermediate gearing cover with new gasket.

Note: Make sure that the sliding sleeve on the hydro-pump input shaft is not displaced.

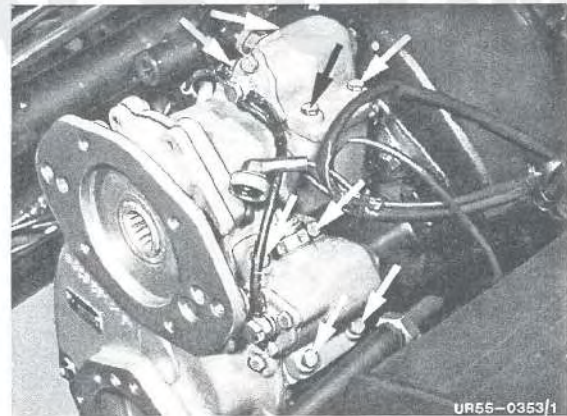


- 14 Fit air lines to the selector cylinders.

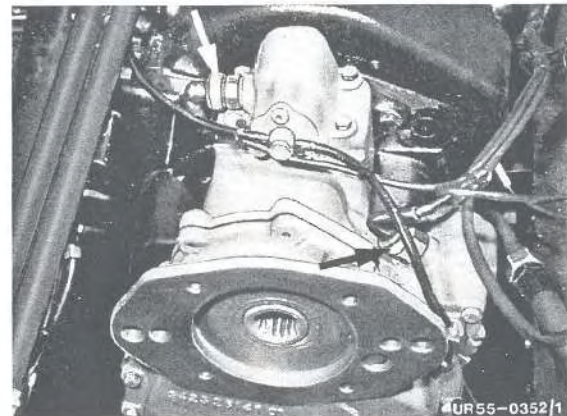


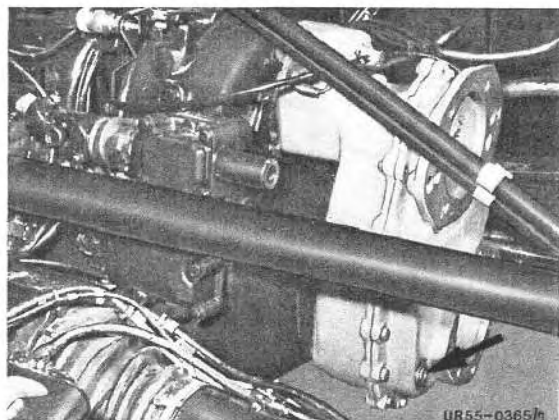
- 15 Coat selector cylinder with sealant No. 1 on mating surface and bolts and fit.

Note: The selector cylinder with the longer selector fork is mounted laterally.



- 16 Fit cable connector to the selector cylinders.





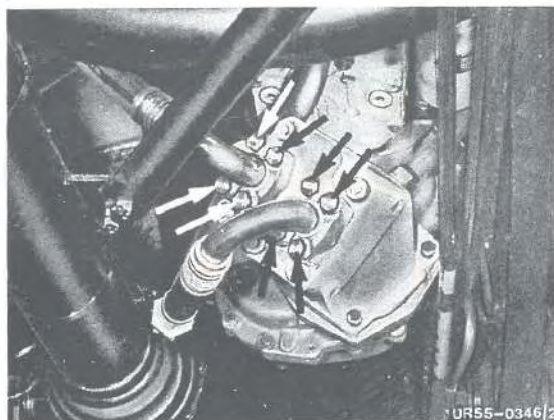
17 Fill with transmission oil.
Capacity, see 1.2/1.

18 Check transmission oil level.

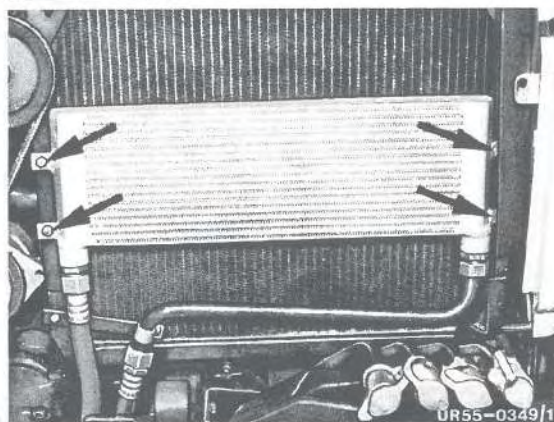
19 Install hydropump,
see 4.1/1.

Removal

- 1 Raise engine bonnet and remove front cover.
- 2 Drain and retain oil from the hydromotor and pump.



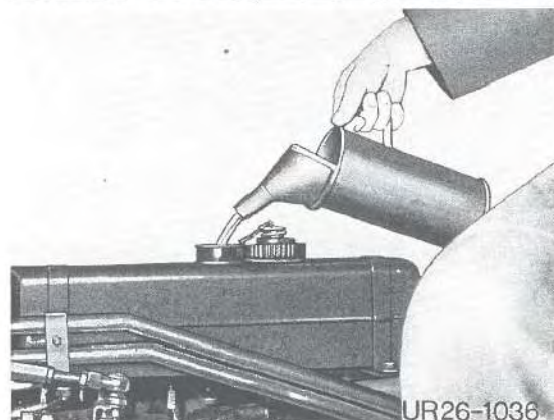
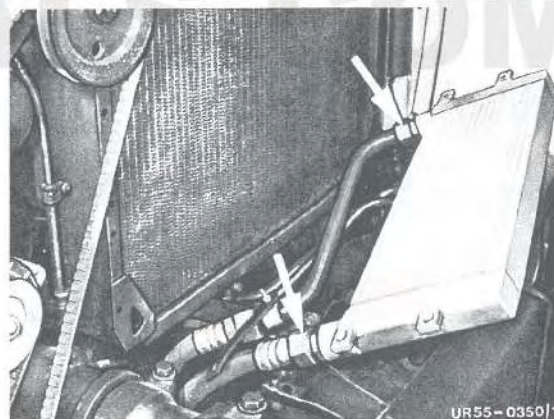
- 3 Remove oil cooler from radiator frame.



- 4 Remove supply and return lines from oil cooler.
- 5 Clean all parts, check and replace if necessary.

Installation

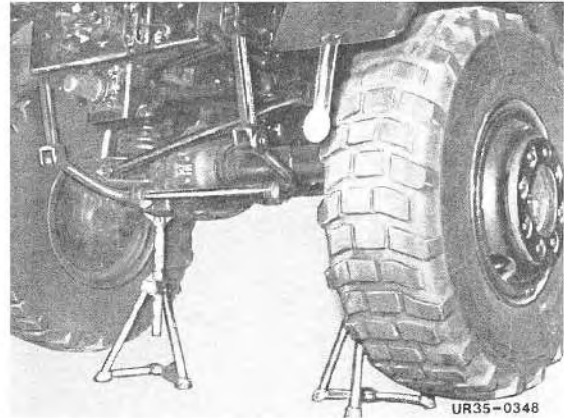
- 1 Fit supply and return lines.
- 2 Install oil cooler to radiator frame.
- 3 Fill expansion tank with oil.
- 4 Bleed hydrostat oil circuit, see 7.1/1.



Bleeding

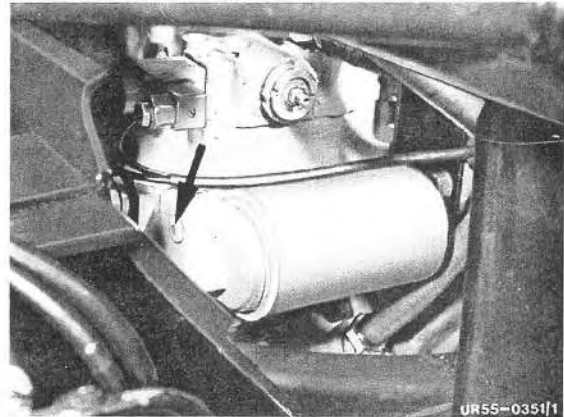
Note: The hydromotor must be able to turn under light loading.

1 Disengage four-wheel drive and support vehicle at rear axle.



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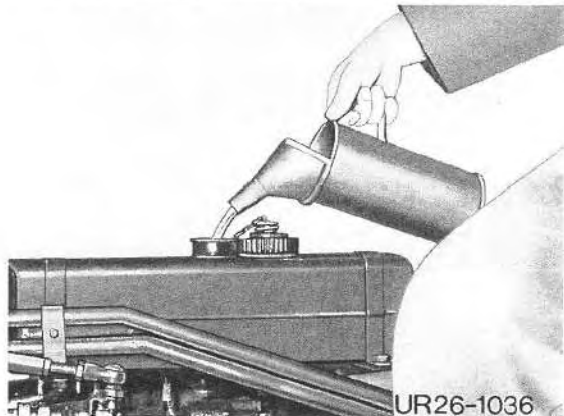
2 Loosen bleed screw on the hydropump filter casing.



UR55-0351/1

3 Fill expansion tank with oil up to maximum level.

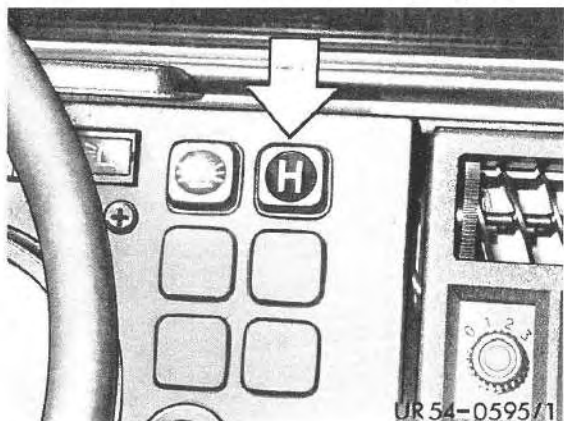
Note: Do not tighten cap.



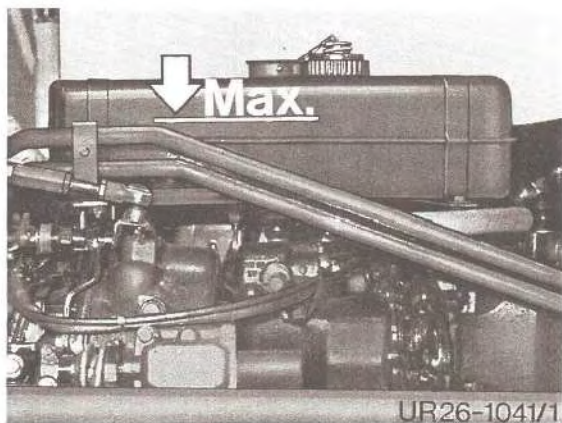
UR26-1036

4 Start engine, engage hydrostat and leave running at idle for 10 sec. — switch off.

5 Wait for two to three minutes, so that the air can escape from the pump casing.

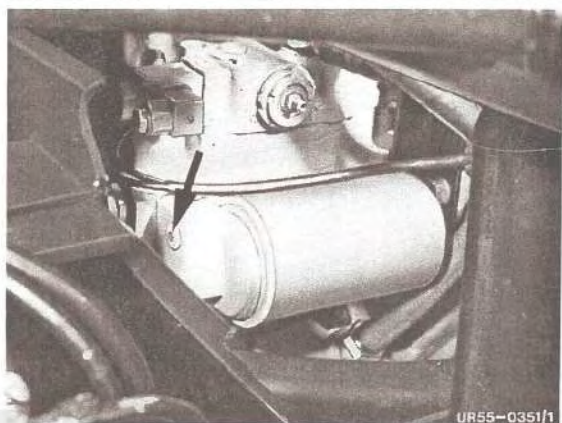


UR 54-0595/1

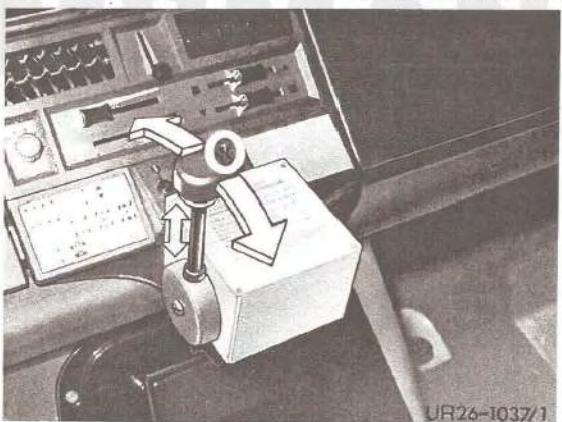


- 6 Repeat points 4 and 5 about 5 times, watching oil level in expansion tank.

Note: The tank is at maximum capacity when 3/4 full.



- 7 Tighten bleed screw on hydropump filter casing when oil flows out.



- 8 Let engine and hydrostat speed increase and make the hydropump deliver its maximum swept volume about 3 times for approx. 30 seconds eachway in both delivery directions.



- 9 Release throttle and warm up with a gradually increasing engine speed and loading, so that the residual air can escape.

- 10 Check oil level in expansion tank and top up if necessary.

Note: The tank is at maximum capacity when 3/4 full.

Survey

Version

Driver's Cab

60.5

Box-Type Body

60.12